

Enhancing Decentralized and Personalized Federated Learning With Topology Construction

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Abstract—The emerging Federated Learning (FL) permits all workers (e.g., mobile devices) to cooperatively train a model using their local data at the network edge. In order to avoid the possible bottleneck of conventional parameter server architecture, the decentralized federated learning (DFL) is developed on the peer-to-peer (P2P) communication. Non-IID issue is a key challenge in FL and will significantly degrade the model training performance. To this end, we propose a personalized solution called TOPFL, in which only parts of the local models (not the entire models) are shared and aggregated. Moreover, considering the limited communication bandwidth on workers, we propose a topology construction algorithm to accelerate the training process. To verify the convergence of the decentralized training framework, we theoretically analyze the impact of the data heterogeneity and topology on the convergence upper bound. Extensive simulation results show that TOPFL can achieve $2.2\times$ speedup when reaching convergence and 5.8% higher test accuracy under the same resource consumption, compared with the baseline solutions.

Index Terms—Personalized federated learning, P2P communication, topology construction, edge computing.

I. INTRODUCTION

RECENTLY, with the rapid development of Internet of Things (IoT) and mobile computing, massive data are generated and distributed at network edge. To mine meaningful knowledge from these data, a typical solution is to gather and analyze the distributed data in centralized data centers, which will incur huge communication costs and potentially lead to privacy leakage [1], [2]. Fortunately, by deploying dedicated hardware at network edge [3], edge computing [4], [5] has emerged recently to process user data and train deep learning models on workers (e.g., edge nodes) that are close to users. This distributed model training paradigm is also known as *Federated learning* (FL) [6],

[7], which protects user privacy by training the models locally and only exposing the models or gradients instead of raw data. A fundamental property of FL is data island [8], [9], [10], where data distributions among workers may vary to a large extent. The non-independent and identically distribution (non-IID) across workers is quite common in real life. For example, cameras located in different places (e.g., a shopping mall or a parking lot) will record different kinds of images (customers or cars). Traditional FL algorithms like FedAvg [6] try to train a uniform model across different workers. However, given the non-IID data, each worker intends to train the model towards a local optimum that fits in its own dataset. The optimization directions of local models may differ greatly, resulting in a significant performance degradation of the aggregated model [11]. To this end, the personalized federated learning (PFL) [12] has received increased attention, which trains a personalized model instead of a uniform model for each worker.

Many PFL works are implemented on the Parameter-Server-based architecture, where the parameter server (PS) usually maintains a global model, and each worker pulls and updates the global model to fit in its local data by using regularized loss functions [13], mixing with the local model [14] or other personalized methods [12], [15], [16]. However, with the increasing scale of participating workers and growing size of deep neural network models, the bandwidth on PS will be eventually saturated by a significant amount of traffic for model transmission. In addition, PS is vulnerable to accidental network connection failure or unexpected external attacks [17], leading to potential single-point failure. Therefore, the decentralized FL is proposed as an alternative. The key idea of decentralized FL is to replace communication with the PS by peer-to-peer communication among multiple workers. In decentralized FL, each worker trains a local model and directly aggregates the models from its neighbors. Without the centralized PS, the decentralized architecture can relax the communication pressure and achieve a better scalability.

Under the decentralized architecture, we are confronted with two main challenges to realize effective model training in PFL. First, the bandwidth of workers is limited and dynamic. In edge computing, workers usually connect to the network through wireless links, e.g., Wi-Fi or mobile cellular networks. These low-speed and time-varying links are likely to suffer high latency when transmitting large-scale models. Second, the computing capacities of workers are heterogeneous. For instance, as depicted in Section V-A, we train a classical neural model ResNet18 on a TX2 device and a laptop for a batch of data

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samples, and find that it separately takes 0.25s and 0.1s on the TX2 device and the laptop. In a heterogeneous FL system, workers with more powerful computing capacity will finish local training faster than others, but the fast workers have to wait for the stragglers (i.e., the slowest workers) for model aggregation. Thus, the training completion time is usually dominated by the stragglers.

Some recent works have made their efforts in proposing efficient solutions to the decentralized PFL problems [18], [19], [20]. The existing solutions can be divided into two categories: fixed topology based and alterable topology based. For example, Dai et al. [21] proposed to employ personalized sparse masks to customize sparse local models on workers, and then aggregate these local models in a fixed topology. The personalized sparse models efficiently coped with the non-IID issue. Whereas, this solution will inherently incur extra costs in calculating and transmitting the masks. In addition, the model divergence between each worker and its fixed neighbors will degrade the training performance [22], [23]. In alterable topology based solutions, the topology is usually periodically reconstructed to reduce the model divergence among workers. For example, Zantedeschi et al. [24] generated a collaboration graph based on the euclidean distances between different local models. After each local training process, they reconstructed the communication topology and restricted the communication to pairs of workers with similar models. But the dynamic network condition and heterogeneous computing capacities of workers might increase the completion time of model training. Both the two types of solutions require each worker to share the entire model with neighbors. However, it is hard for each worker to make good use of both the common knowledge from neighbors and the specific knowledge from its own. As demonstrated in a PS-based work [25], a globally shared network body with many different personalized local heads can achieve significant success in dealing with non-IID data. By aggregating the bodies on the PS and keeping the local heads locally, each worker can not only capture the common knowledge from others but also better accommodate to its own dataset.

Herein, we propose an efficient solution termed TOPFL for decentralized PFL. Inspired by [25], we split an entire model into a personalized sub-model and a base sub-model. Each worker keeps the personalized sub-model (close to the output) locally to benefit from the user-specific dataset, and shares the base sub-model (close to the input) with its neighbors to learn extra common knowledge. In order to reduce the training completion time, we dynamically reconstruct the P2P network topology. Concretely, we attempt to remove slow links in the original topology, therefore the communication burden of workers can be eased to a certain extent, which enables faster communication and reduce the total training completion time. However, how to determine the network topology is challenging. Generally, more links across workers can improve the quality of local models, but the communication overhead is increased. In contrast, the communication overhead can be reduced with fewer links, but the model performance will be degraded. The main contributions of this paper are outlined as follows:

- In order to tackle the non-IID issue in decentralized PFL, we propose a personalized solution called TOPFL, where

workers only share the base sub-model for aggregation and always keep the personalized sub-model locally.

- To adapt to the limited, dynamic network bandwidth and heterogeneous computing capacity, we propose to reconstruct the topology with a greedy algorithm in terms of the available communication and computing resources on workers.
- Extensive experiments are conducted to evaluate the effectiveness and efficiency of TOPFL. The results show that compared with the baselines, TOPFL can complete the training process faster and achieve higher test accuracy.

The rest of this paper is organized as follows. Section II provides the preliminaries of personalized federated learning and problem formulation. The theoretical convergence analysis is presented in Section III. Section IV gives the detailed description of the proposed algorithm. In Section V, the experiments are conducted and the corresponding results are presented. Section VI introduces some related works, and Section VII concludes the paper.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. Decentralized and Personalized FL

We consider a synchronous decentralized federated learning system with a worker set $V = \{v_1, v_2, \dots, v_n\}$, where n is the number of workers. Each worker v_i trains a local model ω_i on its local dataset D_i , and the local loss function is denoted as $f_i(\omega_i) = \mathbb{E}_{\xi \in D_i}[F_i(\omega_i, \xi)]$, where ξ represents a data sample in D_i . The traditional decentralized PFL optimization problem is represented as

$$\min_{\Omega=[\omega_1, \omega_2, \dots, \omega_n]} f(\Omega) = \frac{1}{n} \sum_{i=1}^n f_i(\omega_i). \quad (1)$$

Besides, the decentralized network topology can be modeled as an undirected graph $\mathcal{G}^t = (\mathcal{V}^t, \mathcal{E}^t)$ at the t th training iteration, where the node set $\mathcal{V}^t$ corresponds to the worker set V and the edge set $\mathcal{E}^t$ represents the communication links among workers. We denote the neighbor set of worker v_i as $\mathcal{N}_i^t$. The link between workers v_i and v_j is denoted as $e_{i,j}^t$. The topology graph can be represented as an adjacent matrix $A^t \in \mathbb{N}^{n \times n}$, where the element $a_{i,j}^t$ can be denoted as

$$a_{i,j}^t = a_{j,i}^t = \begin{cases} 1, & e_{i,j}^t \in \mathcal{E}^t, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

The degree matrix U^t of $\mathcal{G}^t$ is a diagonal matrix, and the diagonal element is the number of neighbors of each worker. Then we express the Laplacian matrix as $L^t = U^t - A^t$. According to the spectral graph theory [26], $\lambda_2(L^t) > 0$ if and only if the topology is connected, where λ_2 is the second smallest eigenvalue of matrix L^t .

B. Decentralized and Personalized FL With Partial Model Aggregation

In this section, we introduce our proposed decentralized and personalized federated learning framework TOPFL, where the partial model aggregation is adopted to realize personalized and communication-efficient model training. As shown in Fig. 1,

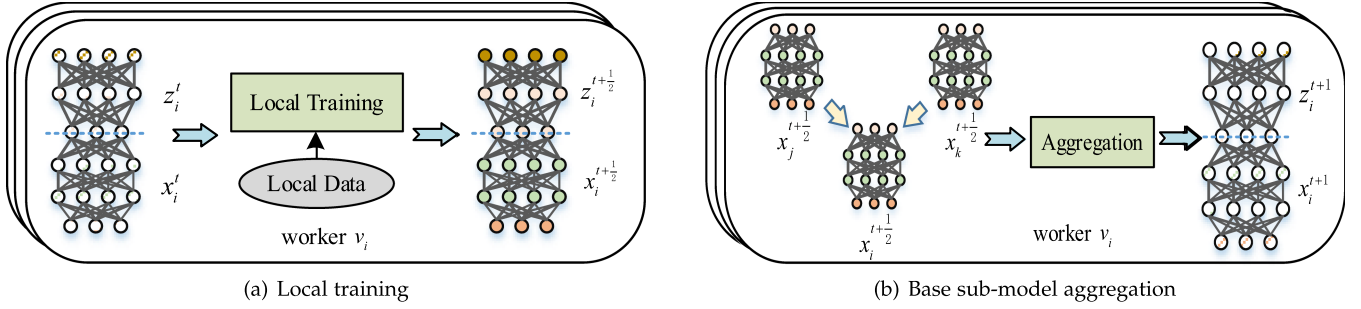


Fig. 1. Training process of our personalized solution TOPFL.

we split the entire local model ω_i into the base sub-model x_i and personalized sub-model z_i . The model parameter vector $\omega_i \in \mathbb{R}^d$ is denoted as $\omega_i = (x_i, z_i)$, where $x_i \in \mathbb{R}^{d_0}$, $z_i \in \mathbb{R}^{d_1}$ and $d_0 + d_1 = d$. The values of d_0 and d_1 depend on the specific splitting layer, which is empirically evaluated in Section V-B. Generally, the base sub-model is responsible for learning common knowledge from others and the personalized sub-model takes charge of learning from the user-specific data. During the training process, each worker trains the entire model locally and then aggregates the trained base sub-model with neighbors. We denote the base sub-model parameters at the t th iteration as $X^t = [x_1^t, x_2^t, \dots, x_n^t] \in \mathbb{R}^{d_0 \times n}$. Similarly, the personalized sub-model parameters are denoted as $Z^t = [z_1^t, z_2^t, \dots, z_n^t] \in \mathbb{R}^{d_1 \times n}$. The optimization problem can be further represented as

$$\min_{X \in \mathbb{R}^{d_0 \times n}, Z \in \mathbb{R}^{d_1 \times n}} f(X, Z) = \frac{1}{n} \sum_{i=1}^n f_i(x_i, z_i), \quad (3)$$

where $X = [x_1, x_2, \dots, x_n]$ and $Z = [z_1, z_2, \dots, z_n]$ are the concatenation of the base sub-model parameters and personalized sub-model parameters, respectively.

We use the mini-batch SGD method to find the optimal solution for the learning problem. At each iteration t , the worker v_i evaluates the stochastic gradient $\nabla F_i(\omega_i^t) = \nabla F_i(x_i^t, z_i^t) = \frac{1}{B} \sum_{j=1}^B \nabla F_i(\omega_i^t, \xi_j^t)$ with a batch of data samples, where B is the batch size. We denote the expectation of the stochastic gradient as $\mathbb{E}_{\xi \in D_i} \nabla F_i(\omega_i^t) = \nabla f_i(\omega_i^t)$.

C. Training Process of TOPFL

To jointly capture the common knowledge of different workers and realize model personalization in decentralized federated learning, each worker keeps its personalized sub-model locally and aggregates the base sub-model with neighbors. The base training process is described in Algorithm 1.

During the local training process of the t th training iteration, the base sub-model x_i^t of worker v_i is updated locally (Line 3)

$$x_i^{t+\frac{1}{2}} = x_i^t - \eta_t \nabla_x F_i(x_i^t, z_i^t), \quad (4)$$

where η_t is the learning rate, and the ∇ operator represents the partial derivative of the sub-model. To facilitate the convergence analysis, we use $x_i^{t+\frac{1}{2}}$ with superscript $t + \frac{1}{2}$ to denote the updated base sub-model after local training at the t th iteration. In fact, we have $\nabla F_i(\omega_i^t) = \nabla F_i(x_i^t, z_i^t) =$

$(\nabla_x F_i(x_i^t, z_i^t), \nabla_z F_i(x_i^t, z_i^t))$. To improve the communication efficiency, we perform the local training in (4) for τ times between each two model aggregations, where τ is a positive integer. After the local updating phase is finished (i.e., $t \bmod \tau = 0$), the coordinator is responsible for determining a new topology and mixing matrix to reduce the communication time, which is described in Section IV. Then each worker v_i will aggregate the base sub-model x_i with others. Under the synchronous decentralized FL, the base sub-model aggregation on worker v_i at the t th training iteration is described as

$$x_i^{t+1} = \sum_{j=1}^n w_{j,i}^t x_j^{t+\frac{1}{2}}, \quad (5)$$

where $w_{j,i}^t \in [0, 1]$ is the element of the mixing matrix $W^t \in \mathbb{R}^{n \times n}$ and represents the aggregation weight of neighbor v_j . If there is no connection between v_j and v_i , $w_{j,i}^t$ is set as zero. The mixing matrix can capture the network topology at the t th iteration. With the mixing matrix W^t , we can unify the local training process in (4) and the model aggregation process in (5) at the t th iteration as

$$X^{t+1} = (X^t - \eta_t G(X^t)) W^t, \quad (6)$$

where $G(X^t)$ is

$$G(X^t) = [\nabla_x F_1(x_1^t, z_1^t), \dots, \nabla_x F_n(x_n^t, z_n^t)]. \quad (7)$$

To unify the representation of the local training and model aggregation processes on worker v_i , we denote the mixing matrix in (6) as an identity matrix at the local training iterations

$$W^t = I, t \bmod \tau \neq 0. \quad (8)$$

Note that $t \bmod \tau = 0$ means each worker will aggregate the base sub-model after local training at the current iteration, and we use the mixing matrix W^t to represent the aggregation weights. $t \bmod \tau \neq 0$ means that each worker only trains the entire model locally, and the mixing matrix W^t is set as an identity matrix to eliminate the influence of the aggregation.

As for the personalized sub-model z_i^t , it is the other part of the local model ω_i^t of worker v_i at the t th training iteration. We train the base sub-model x_i^t and the personalized sub-model z_i^t together as a complete local model ω_i^t and do not treat them separately during the local training phase. That is, we train the personalized sub-model z_i^t for τ times between each two model aggregations. However, the personalized sub-model z_i^t

Algorithm 1: Training Process of TOPFL.

Input: Initial model x^0 , step size η , batch size b , number of local updating iteration τ .
1: Initialize the model $x_i^0 = x^0$, iteration index $t = 0$
2: **for** $t = 0$ to T **do**
(For base sub-model):
3: For each worker v_i : $x_i^{t+\frac{1}{2}} = x_i^t - \eta_t \nabla_x F_i(x_i^t, z_i^t)$;
4: **if** $t \bmod \tau = 0$ **then**
5: The coordinator determines the network topology and mixing matrix for workers;
6: Each worker aggregates the base sub-model in terms of the mixing matrix:
 $x_i^{t+1} = \sum_{j=1}^n w_{j,i}^t x_i^{t+\frac{1}{2}}$
7: **else**
8: $x_i^{t+1} = x_i^{t+\frac{1}{2}}$
(For personalized sub-model):
9: For each worker v_i : $z_i^{t+1} = z_i^t - \eta_t \nabla_z F_i(x_i^t, z_i^t)$
Output: $x_1^t, x_2^t, \dots, x_n^t$

will always stay locally and won't be shared across workers after local training, so as to preserve the personalized feature of the user-specific dataset. We denote the update of z_i^t at the t th iteration as

$$z_i^{t+1} = z_i^t - \eta_t \nabla_z F_i(x_i^t, z_i^t). \quad (9)$$

We define the average base sub-model in the t th training iteration as

$$\bar{x}^t = \frac{1}{n} \sum_{i=1}^n x_i^t. \quad (10)$$

Note that the average model is not available in practice. For decentralized federated learning, the consensus distance is adopted to measure the derivation of the model on each worker from the averaged model. We define the consensus distance as

$$\psi^t = \frac{1}{n} \sum_{i=1}^n \|x_i^t - \bar{x}^t\|^2. \quad (11)$$

According to [27], when the consensus distance is lower than a critical threshold, decentralized training will converge as fast as the centralized counterpart. We denote the threshold as Ψ^t , which is a positive constant.

D. Problem Definition

Edge computing has supported the development of federated learning, but the intrinsic features of edge computing will pose new challenges for efficient model training. The limited and dynamic bandwidth on workers has significant impact on the communication time, and the heterogeneous capacities of workers will lead to extra waiting time. We intend to reduce the model training completion time of each round, which is the combination of τ local updates and one model aggregation.

For the sake of representation, we use the round index k instead of the iteration index t to formulate our optimization problem. Each training round consists of τ iterations. The

amount of computing resource available for each worker is different, and thus the computing time differs among workers. For worker v_i , we denote the computing time at the t' th iteration in round k as $c_i^{k,t'}$. Then the computing time of worker v_i in round k is represented as

$$\mathbf{t}_i^{k,cp} = \sum_{t'=1}^{\tau} c_i^{k,t'}. \quad (12)$$

We reconstruct the topology in every round according to the currently available resources on workers, and thus the neighbor set of each worker is time-varying. The neighbor set of worker v_i in the k th round is $\mathcal{N}_i^k$. After local training, each worker v_i communicates with its neighbors, and the communication time $\mathbf{t}_i^{k,cm}$ of worker v_i in round k is represented as

$$\mathbf{t}_{i,j}^{k,cm} = \frac{s}{b_{ij}^k}, \quad (13)$$

where s is the size of base sub-model and b_{ij}^k is the bandwidth of the link $e_{i,j}$.

The models are trained in a synchronous manner, thus the completion time of each round is determined by the slowest worker. The completion time of each training round k is $\mathbf{t}^k$

$$\mathbf{t}^k = \max_{1 \leq i \leq n} \max_{j \in \mathcal{N}_i^k} (\mathbf{t}_i^{k,cp} + \mathbf{t}_{i,j}^{k,cm}), \quad (14)$$

where $\mathbf{t}_i^{k,cp}$ represents the computing time of worker v_i and $\mathbf{t}_{i,j}^{k,cm}$ communication time between v_i and its neighbor v_j at round k respectively. We formulate the optimization problem as follows:

$$\begin{aligned} \min_{[\mathcal{N}_1^k, \mathcal{N}_2^k, \dots, \mathcal{N}_n^k]} & \left[\mathbf{t}^k = \max_{1 \leq i \leq n} \max_{j \in \mathcal{N}_i^k} (\mathbf{t}_i^{k,cp} + \mathbf{t}_{i,j}^{k,cm}) \right] \\ \text{s.t.} & \begin{cases} \psi^k \leq \Psi^k \\ \lambda_2(L^k) > 0 \\ e_{i,j}^k \in \{0, 1\}, \quad \forall i, j \\ \tau \in \mathbb{N}^+ \end{cases} \end{aligned} \quad (15)$$

The first inequality represents the consensus distance constraint, and we set the predefined threshold empirically according to [27]. The second inequality ensures a connected topology in each round, which is essential to guarantee the training convergence [28]. In the third equation, $e_{i,j}^k$ is a binary variable. $e_{i,j}^k = 1$ means v_i and v_j is connected, and vice versa. The objective is to accelerate the training process of the decentralized PFL with limited and heterogeneous resources. We realize it by periodically adjusting the neighbors of each worker during the model aggregation process.

III. CONVERGENCE ANALYSIS

To analyze the feasibility of the proposed model training mechanism, we prove that TOPFL can achieve a constant convergence bound in this section.

Most objective functions in federated learning are non-convex. Without loss of generality, we analyze the convergence with non-convex optimization. In non-convex optimization, there are usually some common assumptions including

the smoothness assumption and bounded variance assumption [29], [30], [31]. The bounded data heterogeneity assumption can be seen in [32] and [28]. Under decentralized federated learning settings, the mixing matrix assumption and spectral gap assumption are usually significant and standard, which are also introduced in many existing works [33], [34], [35].

Assumption 1. (Smoothness): For all $\omega_1, \omega_2 \in \mathbb{R}^{d_0+d_1}$ and $v_i \in V$, the local objective function $f_i(\cdot)$ is differentiable and L -smooth, i.e.,

$$\|\nabla f_i(\omega_1) - \nabla f_i(\omega_2)\| \leq L\|\omega_1 - \omega_2\|. \quad (16)$$

Assumption 2. (Bounded Variance): For all $v_i \in V$ and $x \in \mathbb{R}^{d_0}, z_i \in \mathbb{R}^{d_1}$, the stochastic gradients $\nabla_x F_i(x, z_i), \nabla_z F_i(x, z_i)$ satisfy

$$\frac{1}{n} \sum_{i=1}^n \mathbb{E}_\xi [\|\nabla_x F_i(x, z_i) - \nabla_x f_i(x, z_i)\|^2] \leq \sigma_1^2, \quad (17)$$

$$\frac{1}{n} \sum_{i=1}^n \mathbb{E}_\xi [\|\nabla_z F_i(x, z_i) - \nabla_z f_i(x, z_i)\|^2] \leq \sigma_2^2, \quad (18)$$

where σ_1, σ_2 are all positive constants.

Assumption 3. (Bounded data heterogeneity): There is a positive constant $\mu > 0$ such that for all $v_i \in V$ and $x \in \mathbb{R}^{d_0}, z_i \in \mathbb{R}^{d_1}$, we have

$$\frac{1}{n} \sum_{i=1}^n \|\nabla f_i(x, z_i)\|^2 \leq \mu \left[\frac{1}{n} \sum_{i=1}^n \|\nabla f_i(x, z_i)\|^2 + \sigma_0^2 \right], \quad (19)$$

where σ_0 is a positive constant related to the data heterogeneity.

Assumption 4. (Mixing Matrix): We assume that the mixing matrix $W \in [0, 1]^{n \times n}$, $(W)_{ij} = w_{ij}^t$ is a symmetric doubly stochastic matrix, that is, $W = (W)^\top$, $W\mathbf{1}_n = \mathbf{1}_n$, and $\mathbf{1}_n^\top W = \mathbf{1}_n^\top$.

Assumption 5. (Spectral gap): Except for the largest eigenvalue of W , the magnitudes of the other eigenvalues are strictly smaller than 1: $|\lambda_n(W)| = 1 > |\lambda_{n-1}(W)| \geq \dots \geq |\lambda_1(W)|$. We define $\rho^t \triangleq \max_{i \in \{2, \dots, n\}} |\lambda_i(W)| = |\lambda_{n-1}(W)|$.

Assumption 6. (Expected Consensus Rate): Under the dynamic topology, the mixing matrix W^t is time-varying, and we have

$$\|X^t W^t - \bar{X}^t\|^2 \leq (1-p)\|X^t - \bar{X}^t\|^2, \quad (20)$$

where $p \in (0, 1]$ is a positive constant related to the spectrum of $(W^t)^\top W^t$ and $\bar{X}^t = \frac{1}{n} X^t \mathbf{1}_n \mathbf{1}_n^\top$ is the average model across all workers.

For ease of representation, we give some definitions below:

Definition 1: At a certain iteration t , we define the average base sub-model $\bar{x}^t$ (a virtual iterative parameter) and average base sub-model gradients h^t as

$$\bar{x}^t = \frac{1}{n} \sum_{i=1}^n x_i^t, \quad (21)$$

$$h^t = \frac{1}{n} \sum_{i=1}^n \nabla_x F_i(x_i^t, z_i^t). \quad (22)$$

Then we have $\bar{x}^t = \bar{x}^{t-1} - \eta_{t-1} h^{t-1}$.

Definition 2: At iteration t , we define the following three types of model parameters:

$$\omega_i^t = ((x_i^t)^\top, (z_i^t)^\top)^\top, \quad (23a)$$

$$\omega_i^* = ((\bar{x}^*)^\top, (z_i^*)^\top)^\top, \quad (23b)$$

$$\bar{\omega}_i^t = ((\bar{x}^t)^\top, (z_i^t)^\top)^\top, \quad (23c)$$

where ω_i^t and ω_i^* represent the individual local model and optimal model on each worker v_i , respectively. $\bar{\omega}_i^t$ denotes that the model consists of an averaged base sub-model and local personalized model on worker v_i .

We first give some common remarks for ease of analysis.

Remark 1: For arbitrary set of n vectors $\{a_i\}_{i=1}^n$, $a_i \in \mathbb{R}^d$, $\|\cdot\|$ is L_2 norm

$$\left\| \sum_{i=1}^n a_i \right\|^2 \leq n \sum_{i=1}^n \|a_i\|^2. \quad (24)$$

Remark 2: Given two vectors $a, b \in \mathbb{R}^d$, $\|\cdot\|$ is L_2 norm

$$\|[a, 0]\|^2 = \|a\|^2 \leq \|[a, b]\|^2. \quad (25)$$

Eq. (25) naturally holds according to the definition of L_2 norm.

Remark 3: Given two vectors $a, b \in \mathbb{R}^d$, and $\langle \cdot \rangle$ represents the inner product of vectors.

$$\langle a, b \rangle = \frac{1}{2} \|a\|^2 + \frac{1}{2} \|b\|^2 - \frac{1}{2} \|a - b\|^2. \quad (26)$$

We then introduce some useful lemmas to facilitate our analysis:

Lemma 4: Under Assumptions 2 and 3, there holds that

$$\begin{aligned} \mathbb{E}_\xi [\|h^t\|^2] &+ \frac{1}{n} \sum_{i=1}^n \mathbb{E}_\xi [\|\nabla_z F_i(x_i^t, z_i^t)\|^2] \\ &\leq \mu \left[\frac{1}{n} \sum_{i=1}^n \|\nabla f_i(\omega_i^t)\|^2 + \sigma_0^2 + \frac{\sigma_1^2}{n} + \sigma_2^2 \right]. \end{aligned} \quad (27)$$

Proof: Note that we can expand $\mathbb{E}[\|h^t\|^2]$ as

$$\begin{aligned} &\mathbb{E} [\|h^t\|^2] \\ &= \mathbb{E} \left[\left\| \frac{1}{n} \sum_{i=1}^n \nabla_x F_i(\omega_i^t) \right\|^2 \right] \\ &\stackrel{(i)}{=} \mathbb{E} \left[\left\| \frac{1}{n} \sum_{i=1}^n (\nabla_x F_i(\omega_i^t) - \nabla_x f_i(\omega_i^t)) \right\|^2 \right] \\ &\quad + \left\| \frac{1}{n} \sum_{i=1}^n \nabla_x f_i(\omega_i^t) \right\|^2 \\ &\stackrel{(ii)}{=} \frac{1}{n^2} \sum_{i=1}^n \mathbb{E} [\|\nabla_x F_i(\omega_i^t) - \nabla_x f_i(\omega_i^t)\|^2] \\ &\quad + \left\| \frac{1}{n} \sum_{i=1}^n \nabla_x f_i(\omega_i^t) \right\|^2 \end{aligned}$$

$$\begin{aligned}
& \stackrel{(iii)}{\leq} \frac{1}{n^2} \sum_{i=1}^n \sigma_1^2 + \left\| \frac{1}{n} \sum_{i=1}^n \nabla_x f_i(\omega_i^t) \right\|^2 \\
& \stackrel{(iv)}{\leq} \frac{1}{n} \sum_{i=1}^n \|\nabla_x f_i(\omega_i^t)\|^2 + \frac{\sigma_1^2}{n},
\end{aligned}$$

where (i) is because that $\nabla_x F_i(\omega_i^t)$ is the unbiased estimation of $\nabla_x f_i(\omega_i^t)$, (ii) is due to the independence of $\nabla_x F_i(\omega_i^t)$ and $\nabla_x f_i(\omega_i^t)$ across different workers, (iii) is by Assumption 2, and (iv) is by (24).

As for the other part in Lemma 4, we expand it as

$$\begin{aligned}
& \frac{1}{n} \sum_{i=1}^n \mathbb{E} [\|\nabla_z F_i(\omega_i^t)\|^2] \\
& = \frac{1}{n} \sum_{i=1}^n \mathbb{E} [\|\nabla_z F_i(\omega_i^t) - \nabla_z f_i(\omega_i^t)\|^2] + \frac{1}{n} \sum_{i=1}^n \|\nabla_z f_i(\omega_i^t)\|^2 \\
& \leq \frac{1}{n} \sum_{i=1}^n \|\nabla_z f_i(\omega_i^t)\|^2 + \sigma_2^2.
\end{aligned}$$

Combining the two inequalities, Lemma 4 is proofed.

Lemma 5: Under Assumption 1, there holds that

$$\begin{aligned}
& -\eta_t \left\langle \frac{1}{n} \sum_{i=1}^n \nabla_x f_i(\bar{\omega}_i^t), \frac{1}{n} \sum_{i=1}^n \nabla_x f_i(\omega_i^t) \right\rangle \\
& - \frac{\eta_t}{n} \sum_{i=1}^n \langle \nabla_z f_i(\bar{\omega}_i^t), \nabla_z f_i(\omega_i^t) \rangle \\
& \leq -\frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\bar{\omega}_i^t) \right\|^2 - \frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\omega_i^t) \right\|^2 \\
& + \frac{\eta_t L}{2} \psi^t, \tag{28}
\end{aligned}$$

where $\psi^t = \frac{1}{n} \sum_{i=1}^n \|x_i^t - \bar{x}^t\|^2$ is the consensus distance of the base sub-models.

Proof: By (26), we have

$$\begin{aligned}
& -\eta_t \left\langle \frac{1}{n} \sum_{i=1}^n \nabla_x f_i(\bar{\omega}_i^t), \frac{1}{n} \sum_{i=1}^n \nabla_x f_i(\omega_i^t) \right\rangle \\
& = -\frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n \nabla_x f_i(\bar{\omega}_i^t) \right\|^2 - \frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n \nabla_x f_i(\omega_i^t) \right\|^2 \\
& + \frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n (\nabla_x f_i(\bar{\omega}_i^t) - \nabla_x f_i(\omega_i^t)) \right\|^2 \\
& \leq -\frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n \nabla_x f_i(\bar{\omega}_i^t) \right\|^2 - \frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n \nabla_x f_i(\omega_i^t) \right\|^2 \\
& + \frac{\eta_t}{2n} \sum_{i=1}^n \|\nabla_x f_i(\bar{\omega}_i^t) - \nabla_x f_i(\omega_i^t)\|^2,
\end{aligned}$$

where the second inequality follows from (24).

Similarly, we have

$$\begin{aligned}
& -\frac{\eta_t}{n} \langle \nabla_z f_i(\bar{\omega}_i^t), \nabla_z f_i(\omega_i^t) \rangle \\
& = -\frac{\eta_t}{2n} \sum_{i=1}^n \|\nabla_z f_i(\bar{\omega}_i^t)\|^2 - \frac{\eta_t}{2n} \sum_{i=1}^n \|\nabla_z f_i(\omega_i^t)\|^2 \\
& + \frac{\eta_t}{2n} \sum_{i=1}^n \|\nabla_z f_i(\bar{\omega}_i^t) - \nabla_z f_i(\omega_i^t)\|^2 \\
& \leq -\frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n \nabla_z f_i(\bar{\omega}_i^t) \right\|^2 - \frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n \nabla_z f_i(\omega_i^t) \right\|^2 \\
& + \frac{\eta_t}{2n} \sum_{i=1}^n \|\nabla_z f_i(\bar{\omega}_i^t) - \nabla_z f_i(\omega_i^t)\|^2.
\end{aligned}$$

Combining the above equations, we have

$$\begin{aligned}
& -\eta_t \left\langle \frac{1}{n} \sum_{i=1}^n \nabla_x f_i(\bar{\omega}_i^t), \frac{1}{n} \sum_{i=1}^n \nabla_x f_i(\omega_i^t) \right\rangle \\
& - \frac{\eta_t}{n} \sum_{i=1}^n \langle \nabla_z f_i(\bar{\omega}_i^t), \nabla_z f_i(\omega_i^t) \rangle \\
& \leq -\frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\bar{\omega}_i^t) \right\|^2 - \frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\omega_i^t) \right\|^2 \\
& + \frac{\eta_t}{2n} \sum_{i=1}^n \|\nabla f_i(\bar{\omega}_i^t) - \nabla f_i(\omega_i^t)\|^2 \\
& \stackrel{(i)}{\leq} -\frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\bar{\omega}_i^t) \right\|^2 - \frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\omega_i^t) \right\|^2 \\
& + \frac{\eta_t L^2}{2n} \sum_{i=1}^n \|w_i^t - w^k\|^2 \\
& = -\frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\bar{\omega}_i^t) \right\|^2 - \frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\omega_i^t) \right\|^2 \\
& + \frac{\eta_t L^2}{2} V^t,
\end{aligned}$$

where (i) is followed by Assumption 1.

Lemma 6: Under the fixed topology with Assumptions 4 and 5, we set $\eta_t = \eta$, then there holds that

$$\begin{aligned}
& \frac{1}{T} \sum_{t=0}^{T-1} \psi^t \leq \eta^2 \sigma_1^2 \left(\frac{1+\rho^2}{1-\rho^2} \tau - 1 \right) \\
& + \eta^2 Q \cdot \frac{1}{T} \sum_{t=0}^{T-1} \frac{1}{n} \sum_{i=1}^n \|\nabla f_i(\omega_i^t)\|^2, \tag{29}
\end{aligned}$$

where $Q = \frac{\tau^2}{1-\rho} \left(\frac{2\rho^2}{1+\rho} + \frac{2\rho}{1-\rho} + \frac{\tau-1}{\tau} \right)$. This upper bound of the consensus distance comes from the results in [36] with the consideration that $\|\nabla f_i(x_i^t)\|^2 \leq \|\nabla f_i(\omega_i^t)\|^2$.

Lemma 7: Under the dynamic topology with Assumptions 4 and 6, we set $\eta_t = \eta$, then the upper bound of consensus

distance is

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \psi^t &\leq \frac{1}{8TL^2} \sum_{t=1}^T \left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\bar{\omega}_i^t) \right\|^2 \\ &\quad + 64 \left(\sigma_0^2 + \frac{12\tau}{p} \right) \eta^2. \end{aligned} \quad (30)$$

This upper bound comes from the analysis in [28] with the consideration that $\|\nabla f_i(\bar{x}^t)\|^2 \leq \|\nabla f_i(\bar{\omega}^t)\|^2$.

Impacted by the dynamic topology, the spectral gap parameter ρ will be time-varying. Thus Lemma 6 can not be directly applied into the convergence analysis under a dynamic topology. To cope with the time-varying topology, we introduce Lemma 7 to bound the consensus distance across workers.

Theorem 8: On the basis of Assumptions 1–5 and Lemmas 4–6, let $\eta_t = \eta$. If η satisfies $\eta L\mu + \mu\eta^2 L^2 Q - 1 \leq 0$, we can obtain the overall convergence bound for the non-convex objective function after T training iterations

$$\begin{aligned} &\frac{1}{T} \sum_{t=1}^T \mathbb{E} \left[\left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\bar{\omega}_i^t) \right\|^2 \right] \\ &\leq \frac{2(f^0 - f^*)}{\eta T} + \frac{\eta L\sigma^2}{n} + \eta^2 L^2 \sigma_0^2 Q \\ &\quad + \eta^2 L^2 \sigma_1^2 \left(\frac{1 + \rho^2}{1 - \rho^2} \tau - 1 \right), \end{aligned} \quad (31)$$

where f^0 is the initial value of objective function.

Proof: According to Assumption 1, we have

$$f_i(\bar{\omega}_i^{t+1}) - f_i(\bar{\omega}_i^t) - \langle \nabla f_i(\bar{\omega}_i^t), \bar{\omega}_i^{t+1} - \bar{\omega}_i^t \rangle \leq \frac{L}{2} \|\bar{\omega}_i^{t+1} - \bar{\omega}_i^t\|^2.$$

Considering that $\bar{\omega}_i^{t+1} - \bar{\omega}_i^t = -\eta_t(h^t + \nabla_z F_i(\omega_i^t))$, we have

$$\begin{aligned} &f_i(\bar{\omega}_i^{t+1}) - f_i(\bar{\omega}_i^t) \\ &\leq -\eta_t \langle \nabla_x F_i(\bar{\omega}_i^t), h^t \rangle - \eta_t \langle \nabla_z F_i(\bar{\omega}_i^t), \nabla_z F_i(\omega_i^t) \rangle \\ &\quad + \frac{\eta_t^2 L}{2} \|h^t\|^2 + \frac{\eta_t^2 L}{2} \|\nabla_z F_i(\omega_i^t)\|^2. \end{aligned}$$

Taking the expectation into consideration, we further obtain that

$$\begin{aligned} &\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n f_i(\bar{\omega}_i^{t+1}) \right] - \frac{1}{n} \sum_{i=1}^n f_i(\bar{\omega}_i^t) \\ &\leq -\eta_t \left\langle \frac{1}{n} \sum_{i=1}^n \nabla_x f_i(\bar{\omega}_i^t), h^t \right\rangle \\ &\quad - \frac{\eta_t}{n} \sum_{i=1}^n \langle \nabla_z f_i(\bar{\omega}_i^t), \nabla_z f_i(\omega_i^t) \rangle \\ &\quad + \frac{\eta_t^2 L}{2} \mathbb{E} [\|h^t\|^2] + \frac{\eta_t^2 L}{2n} \sum_{i=1}^n \mathbb{E} [\|\nabla_z F_i(\omega_i^t)\|^2]. \end{aligned}$$

Then combining Lemmas 4 and 5, we have

$$\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n f_i(\bar{\omega}_i^{t+1}) \right] - \frac{1}{n} \sum_{i=1}^n f_i(\bar{\omega}_i^t)$$

$$\begin{aligned} &\leq \frac{\eta_t}{2} (\eta_t L\mu - 1) \left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\omega_i^t) \right\|^2 \\ &\quad - \frac{\eta_t}{2} \left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\bar{\omega}_i^t) \right\|^2 + \frac{\eta_t L^2}{2} \psi^t + \frac{\eta_t^2 \sigma^2 L}{2n}. \end{aligned}$$

where $\sigma^2 = n\sigma_0^2 + \sigma_1^2 + n\sigma_2^2$. By rearranging the inequality and taking the cumulative average about t , we have

$$\begin{aligned} &\frac{1}{T} \sum_{t=1}^T \mathbb{E} \left[\left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\bar{\omega}_i^t) \right\|^2 \right] \\ &\leq \frac{\eta_t L\mu - 1}{T} \sum_{t=1}^T \left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\omega_i^t) \right\|^2 \\ &\quad + \frac{\eta_t L\sigma^2}{n} + \frac{2\mathbb{E}[f_i(\bar{\omega}_i^0) - f^*]}{\eta_t T} + \frac{L^2}{T} \sum_{t=1}^T \psi^t. \end{aligned}$$

Then combining Lemma 6 and expanding $\frac{1}{T} \sum_{t=1}^T \psi^t$, Theorem 8 is proofed.

Corollary 9: On the basis of Assumptions 1–5 and Lemmas 4–6, if the learning rate is set as $\eta = \frac{\sqrt{n}}{\sqrt{T}}$, the average squared gradient norm after T iterations is bounded by

$$\begin{aligned} &\frac{1}{T} \sum_{t=1}^T \mathbb{E} \left[\left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\bar{\omega}_i^t) \right\|^2 \right] \\ &\leq \frac{2(f^0 - f^*)}{\sqrt{T}n} + \frac{L\sigma^2}{\sqrt{T}n} + \frac{nL^2\sigma_0^2 Q}{T} \\ &\quad + \frac{nL^2\sigma_1^2}{T} \left(\frac{1 + \rho^2}{1 - \rho^2} \tau - 1 \right), \end{aligned} \quad (32)$$

where $Q = \frac{\tau^2}{1-\rho} \left(\frac{2\rho^2}{1+\rho} + \frac{2\rho}{1-\rho} + \frac{\tau-1}{\tau} \right)$.

Corollary 10: Under the dynamic topology with Assumptions 1–4, 6 and Lemmas 4–5, 7, let $\eta_t = \eta$. If η satisfies $\eta \leq \frac{1}{\mu L}$, we can obtain the overall convergence bound for the non-convex objective function after T training iterations

$$\begin{aligned} &\frac{1}{T} \sum_{t=1}^T \mathbb{E} \left[\left\| \frac{1}{n} \sum_{i=1}^n \nabla f_i(\bar{\omega}_i^t) \right\|^2 \right] \\ &\leq \frac{16(f^0 - f^*)}{7\eta T} + \frac{8\eta L\sigma^2}{7n} + \frac{512}{7} \left(\sigma_0^2 + \frac{12\tau}{p} \right) \eta^2 L^2, \end{aligned} \quad (33)$$

where we can set the learning rate η as a decreasing sequence to ensure the upper bound converge to zero.

Proposition 1: If the consensus distance is bounded by

$$\psi^t \leq \frac{1}{Ln} \eta \sigma^2 + \frac{1}{8L^2} \left\| \nabla f(\bar{\omega}^t) \right\|^2, \quad \forall t \leq T, \quad (34)$$

where $\bar{\omega}^t = \frac{1}{n} \sum_{i=1}^n \omega_i^t$. Then in decentralized FL we can set the learning rate $\eta \leq O(\frac{1}{L})$ and recover the convergence rate of PS-based FL.

Proof: As shown in [28], under the given assumptions and setting learning rate as $\eta \leq \frac{1}{4L}$, the iterates in decentralized FL

satisfy

$$\mathbb{E}_{t+1} f(\bar{\omega}^{(t+1)}) \leq f(\bar{\omega}^{(t)}) - \frac{\eta}{4} \left\| \nabla f(\bar{\omega}^{(t)}) \right\|_2^2 + \eta \psi^t + \frac{L}{n} \eta^2 \sigma_0^2.$$

By scaling the consensus distance ψ^t with the inequality in (34), we can derive that

$$\mathbb{E}_{t+1} f(\bar{\omega}^{(t+1)}) \leq f(\bar{\omega}^{(t)}) - \frac{\eta}{8} \left\| \nabla f(\bar{\omega}^{(t)}) \right\|_2^2 + \frac{2L}{n} \eta^2 \sigma_0^2,$$

which matches the standard recursion that one can derive for PS-based FL [37], [38], [39], [40].

In DFL, the convergence rate is usually slower than that in centralized FL due to the impact of communication topology. Theorem 8 ensures the convergence of TOPFL, but can not further accelerate the convergence rate. Fortunately, by imposing a threshold on the consensus distance, Proposition 1 provides a theoretical assurance that the convergence rate of TOPFL can catch up with that of centralized FL.

IV. ALGORITHM DESIGN

In this section, we introduce the training process on each worker and propose an topology reconstruction algorithm to solve the optimization problem in (15). Given the consensus distance constraint, we try to find the optimal communication topology for achieving a shorter completion time of each training round.

A. Estimation of Consensus Distance

In decentralized federated learning, each worker only communicates with its neighbors and the averaged model is not available. The average consensus distance is thus actually agnostic. But we can find an approximate evaluation of the average consensus distance. We expand the average consensus distance ψ^t at the t th training iteration as

$$\begin{aligned} \psi^t &= \frac{1}{n} \sum_{i=1}^n \|x_i^t - \bar{x}^t\|^2 \\ &= \frac{1}{n} \sum_{i=1}^n \|x_i^t - \frac{1}{n} \sum_{j=1}^n x_j^t\|^2 = \frac{1}{n} \sum_{i=1}^n \left\| \frac{1}{n} \sum_{j=1}^n (x_i^t - x_j^t) \right\|^2 \\ &\leq \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \|x_i^t - x_j^t\|^2 = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \psi_{i,j}^t, \end{aligned} \quad (35)$$

where $\psi_{i,j}^t = \|x_i^t - x_j^t\|^2$ is the consensus distance between workers v_i and v_j .

Note that we need to compute the consensus distance before base sub-model aggregation, while the base sub-models x_i^t and x_j^t are both aggregated models. That is, the aggregated base sub-models are agnostic, and we only have access to the base sub-models before aggregation. For ease of representation, we define the base sub-model of worker v_i before aggregation as $\tilde{x}_i^t$, which is the same as $x_i^{t+\frac{1}{2}}$ in Section II-C. We represent the square root of the consensus distance $\psi_{i,j}^t$ with the base sub-models before aggregation as

$$\sqrt{\psi_{i,j}^t} = \|\tilde{x}_i^t - \tilde{x}_j^t\|$$

$$\begin{aligned} &= \|\tilde{x}_i^t + \frac{1}{|\mathcal{N}_i^t| + 1} \sum_{r \in \mathcal{N}_i^t} (\tilde{x}_r^t - \tilde{x}_i^t) \\ &\quad - \tilde{x}_j^t - \frac{1}{|\mathcal{N}_j^t| + 1} \sum_{r \in \mathcal{N}_j^t} (\tilde{x}_r^t - \tilde{x}_j^t)\| \\ &\leq \|\tilde{x}_i^t - \tilde{x}_j^t\| + \frac{1}{|\mathcal{N}_i^t| + 1} \sum_{r \in \mathcal{N}_i^t} \|\tilde{x}_r^t - \tilde{x}_i^t\| \\ &\quad + \frac{1}{|\mathcal{N}_j^t| + 1} \sum_{r \in \mathcal{N}_j^t} \|\tilde{x}_r^t - \tilde{x}_j^t\| \\ &= d_{i,j}^t + \frac{1}{|\mathcal{N}_i^t| + 1} \sum_{r \in \mathcal{N}_i^t} d_{i,r}^t + \frac{1}{|\mathcal{N}_j^t| + 1} \sum_{r \in \mathcal{N}_j^t} d_{j,r}^t, \end{aligned} \quad (36)$$

where $\mathcal{N}_i^t$ denotes the neighbor set of worker v_i at the t th training iteration and $d_{i,j}^t = \|\tilde{x}_i^t - \tilde{x}_j^t\|$ is the model divergence between workers v_i and v_j before aggregation. Different from the consensus distance $\psi_{i,j}^t$ in (35), $d_{i,j}^t$ is available when we decide the topology. For each worker v_i , since it can directly receive the base sub-models from its neighbors, the divergence between worker v_i and its neighbors can be easily derived. However, for two non-adjacent workers v_i and v_j , they can not establish communication directly and obtain the base sub-models of each other. To estimate the divergence between two non-adjacent workers, we introduce an auxiliary worker v_a that connects with both v_i and v_j . We denote the divergence $d_{i,j}^t$ as at iteration t

$$\begin{aligned} d_{i,j}^t &= \|\tilde{x}_i^t - \tilde{x}_j^t\| = \|\tilde{x}_i^t - \tilde{x}_a^t + \tilde{x}_a^t - \tilde{x}_j^t\| \\ &\leq \|\tilde{x}_i^t - \tilde{x}_a^t\| + \|\tilde{x}_a^t - \tilde{x}_j^t\| = d_{i,a}^t + d_{a,j}^t, \end{aligned} \quad (37)$$

where the third step follows the triangle inequality rule. According to the (37), we can further estimate the divergence $d_{i,j}^t$ as long as workers v_i and v_j are in a connected graph. For example, if there is a path between v_1 and v_5 : $v_1 - v_2 - v_4 - v_5$, the divergence $d_{1,5}^t$ can be roughly estimated by

$$d_{1,5}^t \leq d_{1,2}^t + d_{2,5}^t \leq d_{1,2}^t + d_{2,4}^t + d_{4,5}^t. \quad (38)$$

B. Algorithm Description

With the estimation of the average consensus distance ψ^t in Section IV-A, we can rewrite the optimization problem in (15) as

$$\begin{aligned} \min_{[\mathcal{N}_1^t, \mathcal{N}_2^t, \dots, \mathcal{N}_n^t]} & \left[\mathbf{t}^t = \max_{1 \leq i \leq n} \max_{j \in \mathcal{N}_i^t} (\mathbf{t}_{i,j}^{t,cp} + \mathbf{t}_{i,j}^{t,cm}) \right] \\ \text{s.t.} & \begin{cases} t \bmod \tau = 0 \\ \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \psi_{i,j}^t \leq \Psi^t \\ \lambda_2(L^t) > 0 \\ e_{i,j}^t \in \{0, 1\}, \quad \forall i, j \\ \tau \in \mathbb{N}^+ \end{cases} \end{aligned} \quad (39)$$

Based on (39), we propose an efficient solution to solve the optimization problem of network topology construction, which is formally described in Algorithm 2. In fact, it is difficult to

Algorithm 2: TopoCons.

Input: The computing time C^t , the bandwidth B^t , the consensus distance constraint Ψ^t .

- 1: Initial the adjacent matrix $A^t = A_b$, search step $m = \lfloor \sqrt{\sum_{i,j} a_{i,j}^t} \rfloor$.
- 2: **while** True **do**
- 3: Calculate the completion time of each link with current topology according to (12) and (13);
- 4: Find m slowest links and add them into set D^t ;
- 5: **for** each link $e_{i,j} \in D^t$ **do**
- 6: delete $e_{i,j}$ and set $a_{i,j}^t = 0$;
- 7: **if** the topology is not connected **then**
- 8: Recover $e_{i,j}$ and set $a_{i,j}^t = 1$;
- 9: Update the topology matrix A^t and calculate the consensus distance ψ_t according to (35)–(37);
- 10: **if** $\psi_t \leq \Psi_t$ **then**
- 11: Update the neighbor set $[\mathcal{N}_1^t, \mathcal{N}_2^t, \dots, \mathcal{N}_n^t]$ and mixing matrix W^t ;
- 12: $m = \lfloor \sqrt{\sum_{i,j} a_{i,j}^t} \rfloor$;
- 13: **else**
- 14: Recover the topology;
- 15: $m = \lfloor m/2 \rfloor$;
- 16: **if** $m = 1$ **then**
- 17: Break;

Output: W^t

directly solve the problem in (39). Since the topology construction decision is a typically 0-1 integer programming problem [41], which is usually NP-Hard. Meanwhile, the time-varying network conditions and computing capabilities of workers also aggravate the difficulty for this problem. To find the optimal solution of (39), we greedily reconstruct the topology to save more communication time under the consensus distance constraint, thus further minimize the completion time of each round.

Specifically, after finishing the local training process, the decentralized system will record the current computing time C^t of workers and the bandwidth B^t , and then reconstruct the topology and recompute the mixing matrix for base sub-model aggregation (see Algorithm 2). Note that the computing power of each worker and the bandwidth of each link in edge computing system is varying, which means that a fixed network topology can not always minimize the communication time. To solve this challenge, we dynamically reconstruct the network topology under the consensus distance constraint. Specifically, we start from the base topology (i.e., A_b) which includes all available links for P2P communication. In order to search the optimal topology efficiently, we first take a large search step. Concretely, we initialize the search step m as the square root of the number of links in current topology. We denote the completion time of each link as $t_{i,j}^t = \max(t_{i,j}^{t,cp}, t_{j,i}^{t,cp}) + t_{i,j}^{t,cm}$, which includes the computing time and communication time of workers. We greedily delete m slowest links with the topology connectivity constraint (Lines 4-8), because the slow links can incur large communication time and become the system bottleneck. Then

we update the topology and calculate the consensus distance under the topology. If the consensus distance constraint is satisfied, we update the neighbor set $[\mathcal{N}_1^t, \mathcal{N}_2^t, \dots, \mathcal{N}_n^t]$ and mixing matrix W^t (Lines 10-12). Otherwise we recover the topology and reduce the search step by half (Lines 14-15). When the search step is reduced to 1, we stop the algorithm.

V. EXPERIMENTATION AND EVALUATION

In this section, we conduct simulation experiments to demonstrate the effectiveness and efficiency of our proposed DFL solution in resource-constrained edge computing.

A. Experimental Settings

System Configuration: We conduct extensive simulations on an AMAX deep learning workstation equipped with an Intel(R) Xeon(R) Gold 5218R CPU @ 2.10GHz, 8 NVIDIA GeForce RTX 3090 GPUs and 128 GB RAM. The implementation of distributed model training is based on the PyTorch framework [42], and the communication module is implemented by the Python socket library.

The communication time is determined by the model size and network bandwidth. To mimic the heterogeneity and dynamics of P2P networks, we let the network bandwidth fluctuate between 10Mb/s and 50Mb/s [43]. Moreover, we evaluate the computation time by extensive tests on different devices, e.g., laptop, Jetson TX2, Jetson AGX etc.. The measured time of each local updating for training AlexNet on CIFAR10 varies in $[0.1s, 0.2s]$, and it varies in $[0.1s, 0.25s]$ for ResNet18 on CIFAR100, $[0.01s, 0.03s]$ for LeNet on EMNIST and $[0.2s, 0.5s]$ for ResNet34 on IMAGE-100.

Datasets and Models: We employ four typical benchmark datasets, EMNIST, CIFAR10, CIFAR100 and ImageNet to evaluate the performance of our proposed solution. Concretely, As an extended version of the popular MNIST dataset, EMNIST contains 731,668 training samples and 82,587 testing samples from 62 classes (10 digits, 52 characters with lowercase and uppercase). The CIFAR10 dataset consists of 60,000 32×32 color images in 10 classes, with 6,000 images per class. The entire dataset is divided into two parts, 50,000 images for training and 10,000 images for test. Different from CIFAR10, CIFAR100 has 100 classes of color images, and each class contains 500 training images and 100 test images. ImageNet is a popular dataset for visual recognition which consists of 1,281,167 training images, 50,000 validation images and 100,000 test images from 1,000 categories, and each sample from ImageNet is a $224 \times 224 \times 3$ image. To cope with the limited resource on edge devices, we create IMAGE-100, a subset of ImageNet that contains 100 out of 1,000 categories, and each sample is resized with the shape of $144 \times 144 \times 3$.

For EMNIST, we use a simple but classical model LeNet to evaluate the solutions in our experiments. LeNet has 2 convolutional layers and 3 fully-connected layers. The learning rate is set as 0.01 with decay rate of 0.98 [44]. For CIFAR10, we train a well-known DNN model AlexNet with 23.27M parameters. The first 8 layers of AlexNet are convolutional and pooling layers, and the remaining 4 layers are fully-connected. As suggested in

[45], we adopt the SGD optimizer in our experiments and set the initial learning rate as 0.01 with decay rate of 0.965. For CIFAR100, we train another well-known DNN model ResNet18, which consists of 1 input layer, 1 output layer and 4 residual blocks. We adopt the SGD optimizer, and the learning rate is initialized as 0.05 with decay rate of 0.98 [46]. The images in IMAGE-100 have larger size and are more diverse than those in other datasets, and thus we choose a deeper neural network ResNet34 for training. ResNet34 also has 4 residual blocks, an input layer and a output layer, but has deeper residual blocks than ResNet18. The learning rate for ResNet34 is initialized as 0.01 with decay rate of 0.98. Except for some specific experiments, all experiments are by default conducted in a Ring topology with 10 workers, and the batch size is 32.

Non-IID Setting: Non-IID issue is a key and common challenge in federated learning. We conduct our experiments with two representative non-IID data partition settings for the training datasets: (i) *latent Dirichlet allocation (LDA)* and (ii) *skewed label*. The non-IID degree of the datasets is denoted by a positive constant p , which poses different means in the above two non-IID settings. Specifically, in LDA setting, $p\%$ non-IID degree denotes each worker owns $p\%$ samples of a label and $\frac{1-p}{n-1}$ of other labels, where n is the number of workers. While in skewed label setting, $p\%$ non-IID degree denotes each worker lacks $p\%$ labels of data. The larger value of p means higher non-IID degree. Specially, we define $p = 0$ to represent the IID distribution.

In LDA setting, each worker is allocated a proportion of the samples of each label. This setting is common in many federated learning works [47], [48]. When the number of labels equals to the number of workers, e.g., partitioning CIFAR10 with 10 labels to 10 workers, each worker will have $p\%$ samples from a certain label and the remaining samples from this label will be uniformly assigned to the other 9 workers. When the number of labels is larger than to the number of workers, each worker owns more samples from multiple labels. For example, if we partition EMNIST with 62 different labels to 10 workers, there will be 8 workers own $p\%$ samples from 6 different labels and 2 workers be assigned with samples from 7 different labels.

In skewed label setting, each worker lacks data samples with some labels. In this non-IID setting, each worker will lack $p\%$ of labels. For example, for CIFAR10 with 10 different labels on 10 workers, if we set $p = 80$, then each worker will randomly choose 2 labels as their target source to sample their training data. If a label is chosen by multiple workers, its data samples will be uniformly assigned to the corresponding workers.

Like most personalized federated learning works [49], [50], [51], we keep the test dataset distribution the same as the training dataset, which is a significant difference in data partition compared with traditional non-personalized FL solutions.

Baselines: To evaluate the effectiveness of our solution on tackling the non-IID issue, we compare our solution with four representative baselines: (i) DisPFL, (ii) DPCG, (iii) D-PSGD, (iv) PureLocal. Concretely, DisPFL and DPCG are two effective decentralized personalized federated learning solutions. DisPFL allows each worker to own a unique sparse model masked by the personalized tailored mask for better adaptation to their local

data. Then each worker aggregates the model parameters from neighbors. DPCG constructs a cooperative graph to describe the relationships between user personal tasks and leverage it for neighbor selection to train personalized models over P2P networks. D-PSGD is a popular decentralized federated learning solution, where each worker trains its model locally for one iteration and then aggregates the model with neighbors. Considering the communication efficiency, we let each worker train the local model with multiple iterations before model aggregation. As shown in [52], this operation will not decrease the final test accuracy. To evaluate the model training performance on its local data, we let each worker train its model locally for the entire training period in the PureLocal solution and take it as a baseline to show the performance improvement of model aggregation in our solution.

B. Impact of Splitting Layer

As mentioned in Section II-B, we split the entire model into a personalized sub-model and a base sub-model for personalized training, but which layer should be chosen as the splitting layer is still uncertain. In this section, we experimentally determine the specific splitting layer by evaluating its impact on the training performance. We compare the performance of the four models: LeNet on EMNIST, AlexNet on CIFAR10, ResNet18 on CIFAR100 and ResNet34 in IMAGE-100, with different splitting layers. Concretely, we split LeNet and AlexNet at the 1st, 2nd, 3rd, 4th layers and the 1st, 3rd, 5th, 7th layers, respectively. ResNet networks usually have a special neural network structure: residual block, which can help overcome the degradation problem in training very deep networks. We split the model at the end of each block for ResNet networks. Specifically, we split the entire model at the 5th, 9th, 13th, 17th layers for ResNet18 and the 7th, 19th, 27th, 33rd layers for ResNet34. For ease of representation, we use 'block1', 'block2', 'block3' and 'block4' to denote the splitting layers for ResNet18 and ResNet34. The splitting layer with higher index is closer to the output, and the lower index means closer to the input layer. It is worth noting that we do not split the model at the output. To better evaluate the impact of the splitting layer on the performance of our solution, we also compare with the traditional DFL method D-PSGD, where each worker shares the entire local model while aggregating the models. To make a distinction with the traditional DFL method, we call our training methods as partial aggregation methods and the traditional method as complete aggregation method.

The results are shown in Fig. 2, where the legend 'full' represents the traditional complete aggregation method. We find that the partial aggregation methods can always outperform the complete aggregation method on non-IID data. For LeNet, AlexNet and ResNet18, the test accuracy of complete aggregation method is lower than that of all other partial aggregation methods. For ResNet34, the partial aggregation methods achieves higher test accuracy than the complete aggregation method with 'block3' and 'block4'. We find that the partial aggregation methods with higher splitting layers can achieve higher test accuracy, but it is not absolutely. For example, for AlexNet on CIFAR10 in

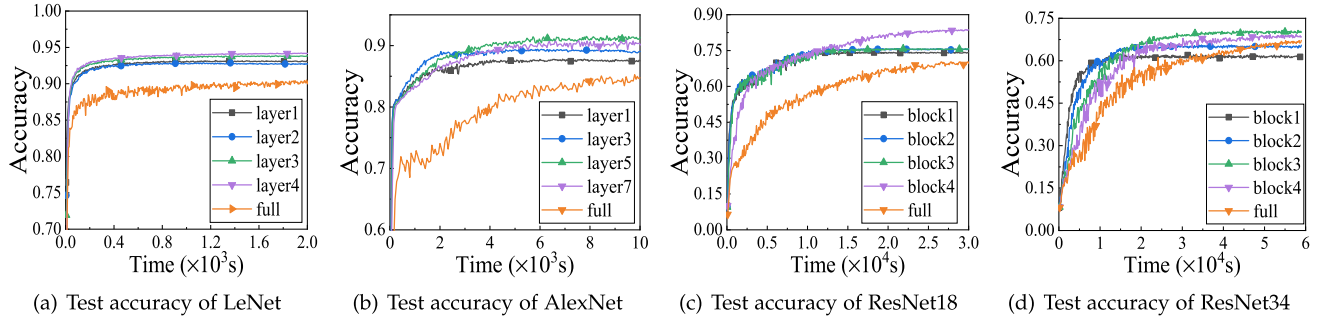


Fig. 2. Training performance given different splitting layers.

Fig. 2(b), the partial aggregation method with the 7th layer achieves higher test accuracy than the methods with the 1st layer and the 3rd layer, while the highest accuracy is achieved by the method with the 5th layer. Similarly, in Fig. 2(d), the partial aggregation method with 'block3' outperforms others for ResNet34 on IMAGE-100.

In conclusion, for a given neural network model, a suitable splitting layer will contribute to achieving a higher final accuracy with non-IID data, and usually it is better to split the entire model at the layers near the last convolutional layer. In the follows, we experimentally set the splitting layer as the 4th layer for LeNet, the 5th layer for AlexNet, the 4th block for ResNet18 and the 3rd block for ResNet34.

C. Impact of Different Number of Workers

In order to evaluate the relationship between the training performance and the number of workers, we also conduct simulations with varying number of workers (from 10 to 30). With the increase in the number of workers, a natural way to partition the training data is to evenly distribute the total training samples among more workers. However, this would result in a reduction of the training samples on each worker, which could degrade the model training performance on each worker. To explore the impact of the amount of training samples, we adopt two different data partition ways: (i) fixed total training samples and (ii) fixed local training samples. In the first way, we keep the amount of total training samples fixed, and the amount of training samples for each worker will become fewer with the increasing number of workers. In the second way, we ensure the amount of training samples on each worker fixed when the number of workers changes by modifying the original data sampling strategy from *sampling without replacement* to *sampling with replacement* [53].

We conduct experiments with TOPFL and D-PSGD on EMNIST and CIFAR10, and the final test accuracy is shown in Figs. 3 and 4. The data distribution is LDA setting with 80% non-IID degree. We notice that TOPFL can always achieve a higher test accuracy compared with D-PSGD and the gap of the two solutions stays stable with varying number of workers, which indicates that the system scale has little impact on the superiority of TOPFL. For example, TOPFL achieves 5.1% higher test accuracy than D-PSGD when the number of workers

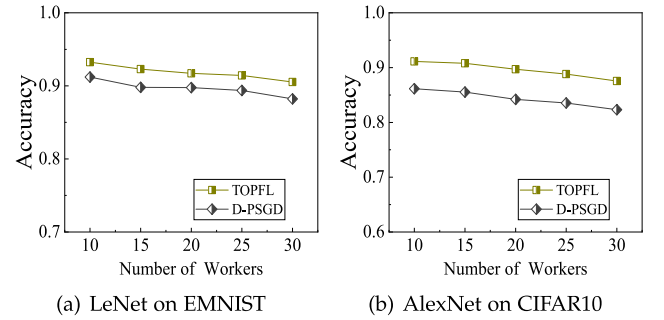


Fig. 3. Training performance on different number of workers with fixed total training samples.

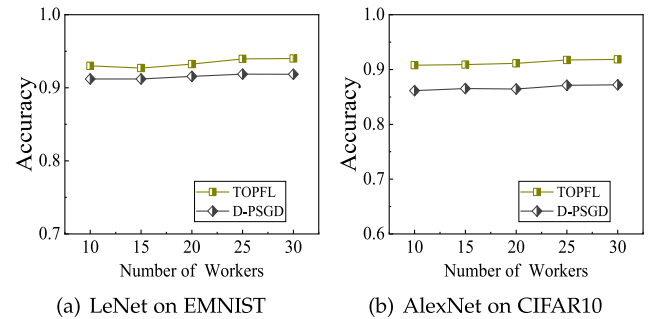


Fig. 4. Training performance on different number of workers with fixed local training samples.

is 10, and the gaps of the two solutions are 5.5% and 5.2% when the numbers of workers are 20 and 30, respectively. As the number of workers increases, the results become different for the two data partition ways. In the fixed total training samples way, the test accuracy of both TOPFL and D-PSGD shows an overall downward trend. Generally, more workers means less trainable samples on each worker, and the available trainable samples on each worker obviously has an impact on its training performance. In addition, more workers involved in training means there are more different optimization directions when aggregating the local models, which results in larger gradient variance and final accuracy degradation. However, when we eliminate the impact of available training samples in the fixed local training samples way, the training performance starts to

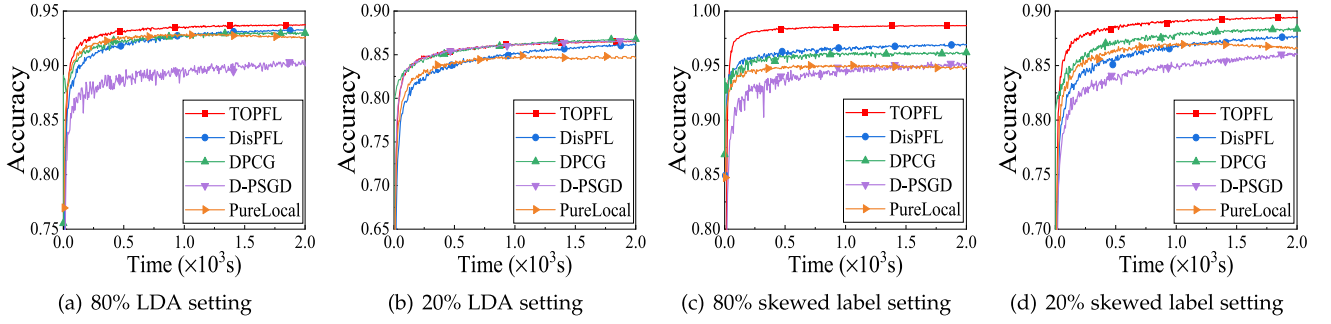


Fig. 5. Test accuracy of LeNet on EMNIST.

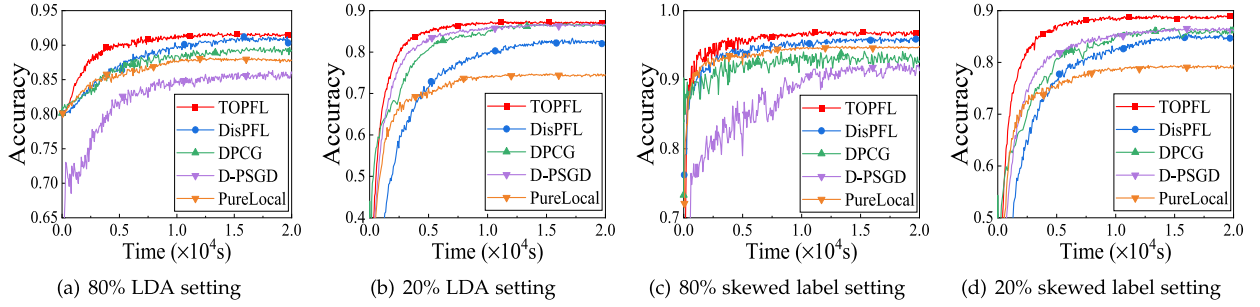


Fig. 6. Test accuracy of AlexNet on CIFAR10.

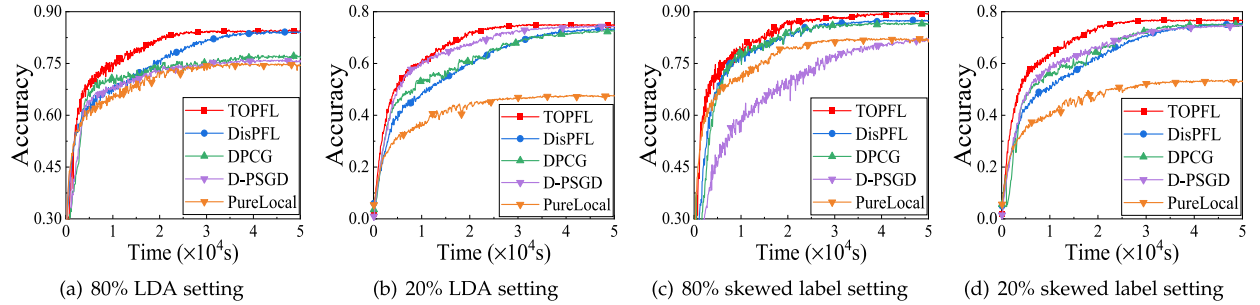


Fig. 7. Test accuracy of ResNet18 on CIFAR100.

slowly improve as the number of nodes increases. This result demonstrates the benefits of the increasing number of workers on the decentralized federated learning.

The results show that the test accuracy gap between TOPFL and D-PSGD is stable with the increasing of the number of workers. On the other hand, more workers in our simulation will increase the memory consumption of GPUs and cause multiplication of simulation completion time. To better present the efficiency of TOPFL and alleviate the system pressure, we set the number of worker as 10 for the other experiments in our paper.

D. Performance Comparison with Baselines

We compare the performance of our TOPFL and other baselines on four benchmark datasets: EMNIST, CIFAR10, CIFAR100 and IMAGE-100 and train four typical models (i.e.,

LeNet, AlexNet, ResNet18 and ResNet34). To further evaluate the effectiveness of TOPFL on non-IID data, we conduct our experiments with two representative non-IID settings (i.e., LDA setting and skewed label setting) which are introduced in Section V-A. We set the non-IID degree as 80% and 20% to respectively represent high and low non-IID degree of each non-IID setting.

The results are summarized in Figs. 5 ~ 8. We observe that TOPFL generally performs better than other baselines with different non-IID degrees. For example, with 80% LDA setting, TOPFL achieves about 91.5% test accuracy with ResNet34 on IMAGE-100, which is 5.8% higher than that of D-PSGD. The personalized baselines DPCG and DisPFL respectively achieve 89.2% and 90.7% test accuracy, which are higher than D-PSGD but lower than TOPFL. With low non-IID degree, TOPFL also achieves the highest test accuracy. As in Fig. 8(b), TOPFL achieves 70.4% test accuracy with 20% LDA setting, which is

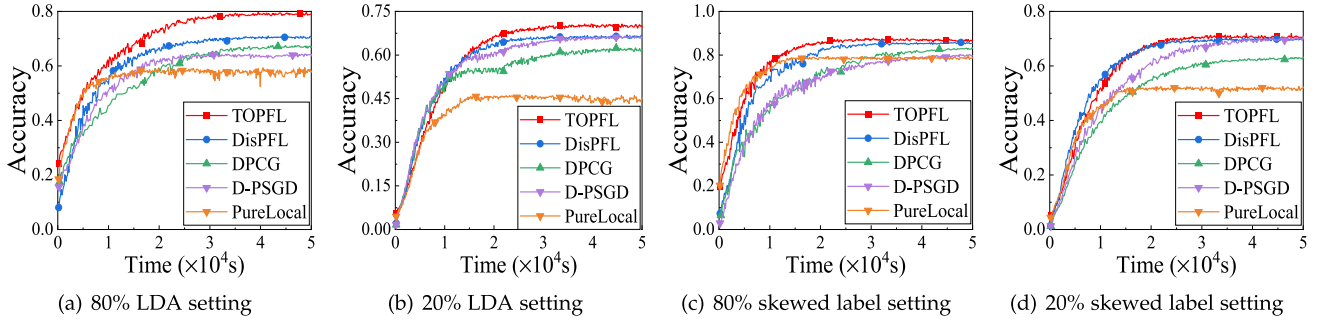


Fig. 8. Test accuracy of ResNet34 on IMAGE-100.

much higher than other baselines. With high non-IID degree, the personalized solutions usually achieve higher test accuracy compared with D-PSGD. As shown in Fig. 5(a), the final accuracy of D-PSGD is 2.8% lower than that of DPCG, 3.1% lower than that of DisPFL and 3.6% lower than that of TOPFL. The traditional federated learning solution will perform worse than the simplest PureLocal solution under high non-IID degree. For example, PureLocal achieves 2.2% higher test accuracy than D-PSGD in Fig. 6(a) and 3.8% higher test accuracy than D-PSGD in Fig. 6(c). This performance degradation means that the model aggregation will increase the updating inconsistency and reduce the final accuracy with high non-IID degree. When the non-IID degree is low, the personalized baselines DisPFL and DPCG sometimes perform not as good as the traditional federated learning solution D-PSGD, while TOPFL always achieves the highest test accuracy. For example, in Fig. 6(b), D-PSGD and TOPFL achieve a similar and the highest test accuracy compared with DisPFL and DPCG. Moreover, in Fig. 6(d), TOPFL achieves the highest test accuracy than others and D-PSGD achieves the second highest test accuracy, while DisPFL and DPCG fail to catch up with D-PSGD. The reason is that both DisPFL and DPCG only take the advantage of knowledge of the specialized data on each worker to improve their training performance, but ignore the common knowledge of other neighbors. Given IID or low-degree non-IID, model aggregation on each worker can utilize the information from others and help improve the training performance, which is validated in many existing decentralized federated learning works [54], [55], [56]. TOPFL has both base sub-model and personalized sub-model. The aggregation of base sub-model can preserve the benefits from others, and the personalized sub-model can help each worker better adapt to its own data. Thus TOPFL can take the advantages from both the specialized data on itself and the data from others with different distributions.

On the other hand, TOPFL can complete training faster than other baselines. For example, in Fig. 6(a), TOPFL consumes about 7,000s to achieve the highest test accuracy, while DisPFL needs 15,500s to reach convergence and achieve a similar test accuracy with TOPFL. In other words, TOPFL achieves $2.2\times$ speedup compared with DisPFL when achieving similar test accuracy. Other baselines like D-PSGD consumes about 15,000s to reach convergence and fails to achieve the same test accuracy with TOPFL.

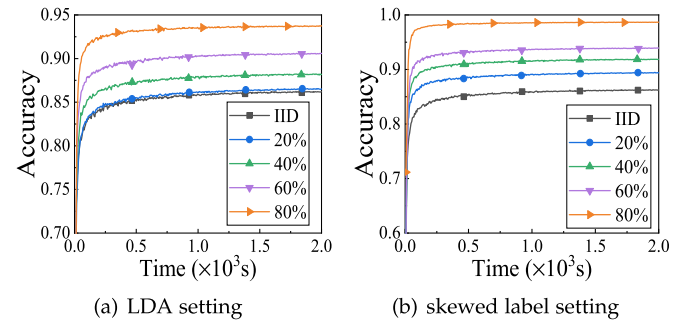


Fig. 9. LeNet on different non-IID degrees.

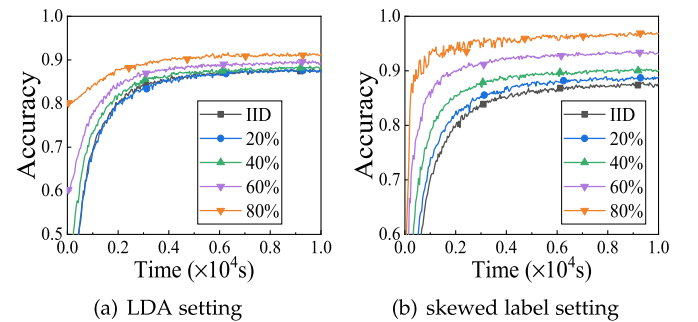


Fig. 10. AlexNet on different non-IID degrees.

E. Impact of Non-IID Data Degree

The non-IID issue, which significantly degrades the model performance, is a key challenge in federated learning. To further evaluate the impact of non-IID degree on TOPFL, we take two non-IID settings (i.e., LDA setting and skewed label setting) into considerations and set the non-IID degree respectively as 20%, 40%, 60% and 80%. We also conduct our experiments on the IID data for comparison.

Figs. 9~12 present the final test accuracy of different models with different non-IID degrees. With the increasing of the non-IID degree, TOPFL achieves higher final test accuracy. For example, as in Fig. 9(a) and (b), TOPFL with 80% non-IID degree achieves the highest test accuracy in both LDA and skewed label settings. It is common for PFL solutions that higher non-IID degree will bring a better test accuracy. Because high

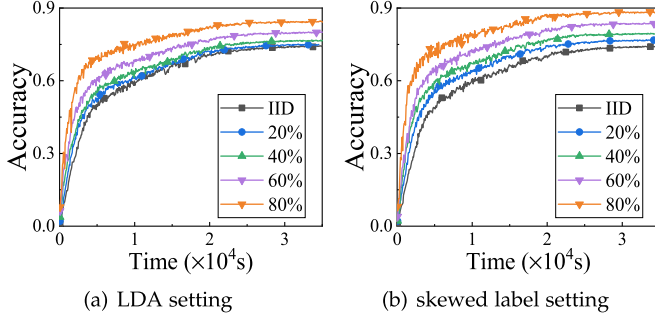


Fig. 11. ResNet18 on different non-IID degrees.

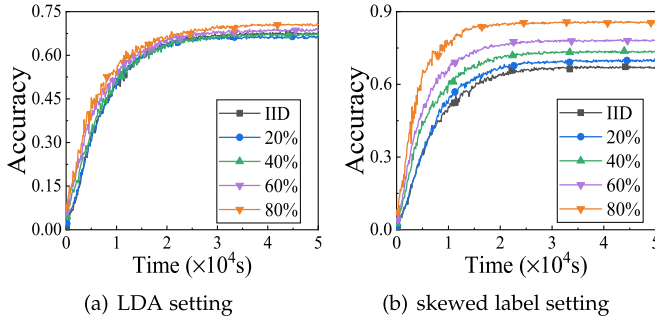


Fig. 12. ResNet34 on different non-IID degrees.

non-IID degree usually means that, for each worker, some labels will account for the majority of its own dataset. In such cases, it will be easier to label these data samples correctly. On the other hand, the test dataset usually follows the same distribution with the training datasets in PFL. The personalized model trained by dataset with high non-IID degree can thus achieve a high test accuracy on the test dataset which is also specialized. The performance gap between TOPFL with different non-IID degrees is more huge in skewed label setting than that in LDA setting. For example, in Fig. 12, the test accuracy gap between TOPFL with 80% non-IID and TOPFL with IID in skewed label setting is 25%, while the gap between them in LDA setting is only about 6%. This phenomenon suggests that the skewed label setting is actually more skewed than LDA setting.

F. Impact of the Computation Method of Consensus Distance

To explore the effectiveness and impacts of our consensus distance evaluation, we respectively conduct experiments with precisely computed consensus distance and estimated consensus distance. To get the precise value of consensus distance in our decentralized system, we let the coordinator collect all the local models after local training and computes the average model in each round. The results are shown in Fig. 13, where “w/ AvgModel” represents the experiments with precisely computed consensus distance based on the average model, “w/o AvgModel” represents the experiments estimating consensus distance without the average model.

We can observe that the test accuracy gap between “w/ AvgModel” and “w/o AvgModel” stays relatively small. The experiments with estimated consensus distance can always achieve similar test accuracy as those with precisely computed consensus distance. This result shows that our method TOPFL successfully achieves a high performance with estimated consensus distance.

VI. RELATED WORK

A. PS-Based PFL

Recently, model personalization is proposed as an effective solution for non-IID issue. A dominate approach in PS-based PFL is local fine-tuning, where each worker receives a global model from PS and then use its own local data to fine-tune it by several local training iterations. For example, Arivazhagan et al. [57] and Collins et al. [25] propose to train global base layers on the PS, and the workers train the remaining personalized layers. On the contrary, Liang et al. [58] proposed a representation learning solution for FL, where they attempt to learn many different local representations (i.e., base sub-models) and a single global head (i.e., personalized sub-model). However, the PS plays an essential role, which is responsible for maintaining a global base model or global personalized model, in these solutions. Therefore, these solutions fail to show their effectiveness in the decentralized FL architecture. In addition, they did not consider which layer should be chosen to split the entire model into two parts, and the splitting layer is usually fixed as the last convolutional layer. However, according to our experiment results in Section V-B, the best splitting layer is not always between the convolutional layers and fully-connected layers.

B. Decentralized PFL

Considering the high communication pressure and high vulnerability to the single-node failure on the PS, some recent works are focusing on the decentralized PFL solutions. Most current decentralized PFL solutions utilize neighbor selection (i.e., topology reconstruction) for workers to train personalized models. Usually the metrics of neighbor selection consists of loss or model distance. Concretely, Onoszko et al. [59] developed PENS in a decentralized networks with non-IID local data. In PENS, each worker tested the models of other workers with its own local data, and it regarded the worker with similar loss as its neighbor. Zantedeschi et al. [24] leveraged a collaboration graph to describe the relationships between workers. The collaboration graph is updated based on the euclidean distances between neighbor models and the local model, and the neighbors with smaller euclidean distances are selected to enhance the personalization performance. Nevertheless, for a resource-constrained worker, these decentralized PFL solutions will introduce unbearable computational cost by evaluating the training loss of neighbors models or computing the euclidean distances. To tackle the data heterogeneity and computing capacity heterogeneity problem together in decentralized PFL, Dai et al. [21] let workers own their personalized unique sparse models masked by the customized mask, which allows them to better

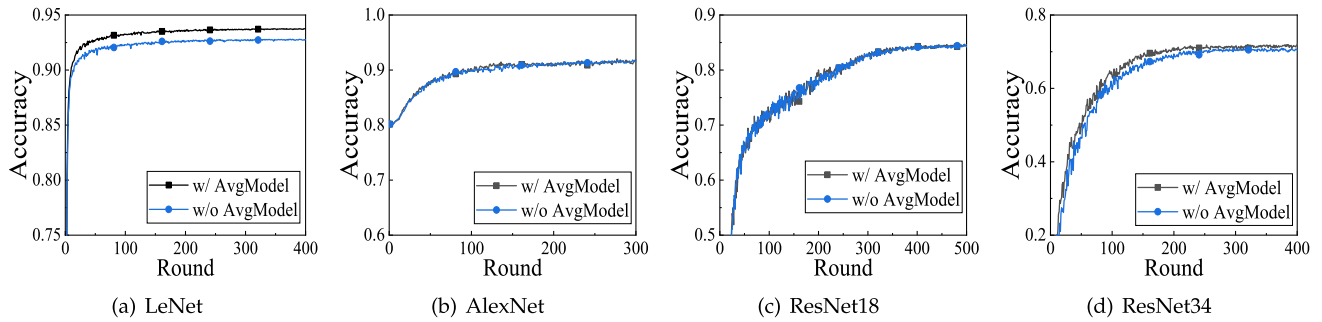


Fig. 13. Impact of average model for consensus distance computation.

adapt to their local data. However, it will lead to extra computing and communication overhead due to the personalized masks.

VII. CONCLUSION

Non-IID issue is a key challenge in FL. In this paper, we introduce a personalized solution called TOPFL to tackle the non-IID issue in decentralized PFL, which split the entire model into base model and personalized model. On the other hand, the communication bandwidth on workers is usually limited and dynamically, which also challenges the efficiency of the decentralized FL approaches. Herein, we reconstruct the network topology to accelerate the training process. we then theoretically analyze the impact of the data heterogeneity and topology on the convergence of the decentralized training framework. Extensive simulation results show that compared with the baseline solutions, TOPFL can achieve $2.2\times$ speedup when reaching convergence and 5.8% higher test accuracy under the same resource consumption.

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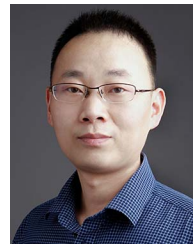
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